

No. of Printed Pages : 4
Roll No.

220746

4th Sem / Civil

Subject : Irrigation Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is a non-recording rainguage?
- a) Floating type rainguage
 - b) Tipping bucket type rainguage
 - c) Simon's rainguage
 - d) Steven's weighing type rainguage
- Q.2 A canal which is aligned at right angles to the contour is called _____
- a) Contour canal
 - b) Watershed canal
 - c) Branch canal
 - d) Side slope canal
- Q.3 Arepelling groyne is aligned _____
- a) Pointing up-stream
 - b) Perpendicular to bank
 - c) Pointing Down -stream
 - d) Parallel to bank

- Q.4 A land is water is water-logged_____
- a) When the permanent wilting point is reached
 - b) When gravity drainage has ceased
 - c) When capillary fringe reaches the root zone of plants
 - d) None of these
- Q.5 The depth of water required to bring the soil moisture content of a given soil upto its field capacity is called_____
- a) Equivalent moisture
 - b) Hygroscopic water
 - c) Soil moisture deficiency
 - d) None of these
- Q.6 A river training work is generally required when the river is_____
- a) Aggrading type b) Meandering type
 - c) Degrading type d) Both (a) and (b)

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 During the growth period of the crop, the quantity of water supplied to a particular crop, giving maximum yield of the crop, is called _____(minimum moisture content / Optimum moisture content)

- Q.8 When the land becomes water logged, the yield of crops in that area _____ (Increases/ Reduces)
- Q.9 In major projects, the area irrigated is more than _____ are multi-purpose in nature. (2000 Hectares / 10000 Hectares)
- Q.10 A curved masonry or concrete dam, convex upstream, that depends principally on the arch for its stability is known as _____ (Arched dam / Masonry dam)
- Q.11 An _____ is provided in the bank of canal to admit small drainage into it. (Inlet / Outlet)
- Q.12 After sowing the crop, the first water applied to the crops is known as _____ water. (Delta/ Kor Water)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Enlist the name of various types of earthen dam. Explain any one of them.
- Q.14 Describe any five various factors causing the water-logging.
- Q.15 Describe the working principle of an automatic rain gauge.
- Q.16 Find the delta for sugarcane when its duty is 730 hectares/ cumec on the field and the base period of the crop being 110 days.

- Q.17 Write a short note on spillways and energy dissipators.
- Q.18 Describe the process of rain water harvesting using “Retention Ponds” and also write its advantages.
- Q.19 Enlist any five difference between weir and barrage.
- Q.20 Enlist the necessity of providing the groynes in river training works.
- Q.21 Write any five factors affecting surface runoff.
- Q.22 Describe any five advantages of Canal Lining.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain water Logging with their five causes effects.
- Q.24 Explain the elementary profile of a gravity dam with the help of diagram.
- Q.25 Explain the method of drip irrigation with the help of diagram and write its advantages.